Exploring microRNA-mRNA Relationships Mediated by Transcription Factors in Cancer

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1 INTRODUCTION

MicroRNAs (miRNAs) regulate **mRNA** expression, and their dysregulation contributes to cancer. **Transcription factors (TFs)**, essential for controlling gene networks, are also targeted by miRNAs.

Traditional correlation analyses often overlook indirect **miRNA-mRNA relationships, regulated by TFs**.

Here, we used **The Cancer Genome Atlas (TCGA) melanoma (SKCM) and breast cancer (BRCA)** datasets (https://www.cancer.gov/tcga), and identified significant miRNA–mRNA correlations. By integrating **known TFs** and **experimentally validated miRNA–TF interactions**, we discovered miRNAs whose correlated genes, that are not direct targets of that miRNA, were significantly enriched in the TF targets set, revealing many indirect miRNA–mRNA relationships potentially mediated by TFs.

2 METHODS

TCGA SKCM Data
452 miRNAs
473 mRNAs

TCGA BRCA Data
1207 miRNAs
1231 mRNAS

A

Preprocessing
- Retain only **protein coding genes** and **matched** miRNA-mRNA samples
- Remove non-expressed and low coverage (<40% samples) features
- Log-transformation

B

Correlation Analysis
Compute pairwise **Spearman correlations;** retain significant (FDR < 0.05) miRNA-mRNA pairs.

C

Identify miRNA-TF interactions
- Downloaded ~1600 known TFs (Lambert et al[1])
- Retain miRNAs targeting at least one TF
- Retrieve miRNA–target interactions using **multiMiR**[2] package validated by experimental methods

D

Identify TF target genes
From ChIP-Atlas, get TF's target genes with peaks within ±5 kb of TSS and in ≥50% cancer cell lines

E

Over-representation Analysis using clusterProfiler[3]
For each miRNA–TF pair, remove direct miRNA targets and test remaining correlated genes for enrichment among TF-target genes (FDR < 0.05)

3 RESULTS

Cohort	Matched Samples	mRNAs	miRNAs	Significant Correlations
SKCM	451	17,384	779	**1,991,909** positive correlations **1,115,724** negative correlations
BRCA	1,198	17,638	572	**1,846,380** positive correlations **2,503,820** negative correlations

Among SKCM samples, we identified **21 miRNAs** that had significant overlap between miRNA's negatively correlated genes and TF targets, and **45 miRNAs** had significant overlap between miR's positively correlated genes and TF targets. For BRCA samples, these numbers were **54 and 57**, respectively (Figure 1).

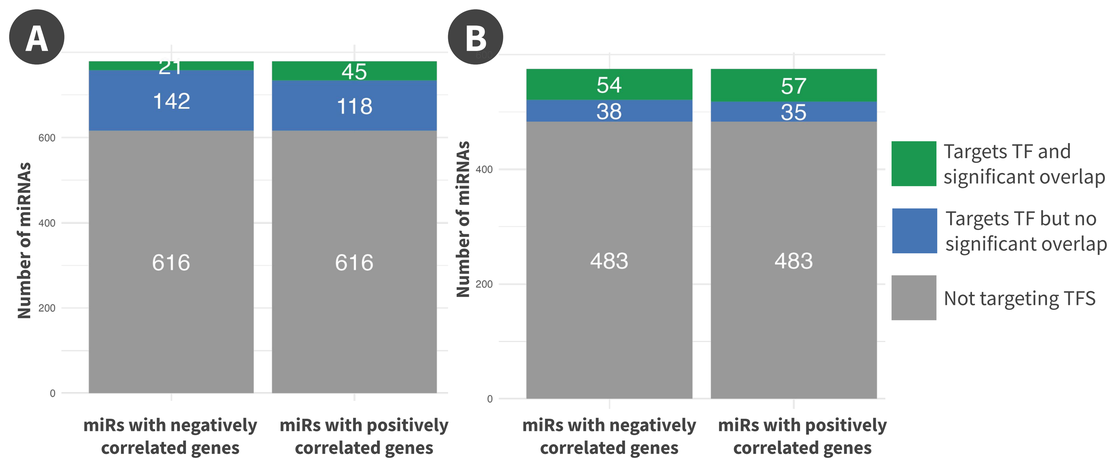


Figure 1. Distribution of miRNAs based on TF interactions and ORA significance. Proportions of all miRNAs that do not target TFs (grey) or target at least one TF with significant (green) of no significant (blue) overlap between miRNA-correlated and TF target genes for (A) SKCM samples, and (B)BRCA samples.

- Since many of the miRNAs target multiple TFs, further ORA revealed significant overlap (FDR < 0.05) between miRNA-correlated genes and TF targets was found for **90 miRNA-TF pairs** among positively correlated genes **31 miRNA-TF pairs** among negatively correlated genes in SKCM; in BRCA data, there were **388 and 300 miRNA-TF pairs** respectively.
- To highlight potentially biologically relevant miRNA–TF relationships, we plotted the top correlated gene from each significant overlapping gene set representing the strongest signal per miRNA–TF pair (Figure 2).

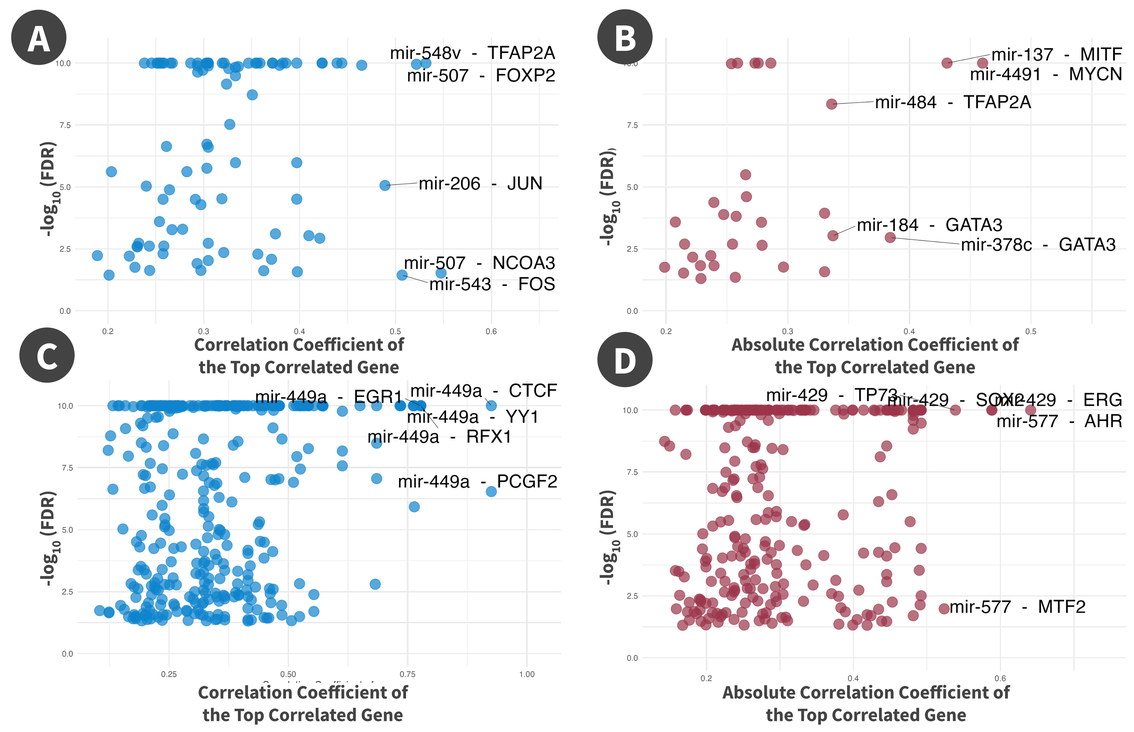


Figure 2. Top correlated gene for each miRNA–TF pair with significant overlap between TF targets and miRNA-correlated genes. (A) positively correlated genes for SKCM samples, (B) negatively correlated genes for SKCM samples, (C) positively correlated genes for BRCA samples, (D) negatively correlated genes for BRCA samples. The top five miRNA–TF pairs with the strongest gene-level correlation and most significant target overlap are annotated.

4 CONCLUSION

- Many miRNA–TF pairs in our study showed significant overlap between miRNA-correlated genes and TF targets, and the **lack of dependence on expression level** suggests that miRNAs' impact on cancer is defined more by its network connectivity than by its abundance.
- For example, **miR-449a in all top 5 miRNA–TF pairs in positive correlations in BRCA** (Figure 2C) — is downregulated in breast cancer and suppresses proliferation, migration, and metastasis by targeting PLAGL2[4].
- Although miRNAs have distinct targets, their **many-to-many relationships** and **miRNA cooperativity** may complicate direct inference. Nevertheless, understanding these layered regulatory networks is key to identify functional interactions and potential biomarkers.
- Future work may include **experimentally validating top miR–TF–mRNA triplets** and integrate clinical data to link networks with patient outcomes.

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